Family Name _____

First Name _____

Student Number |__|__|__|__|__|__|__|__|

Venue _____

Seat Number _____

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Computer Science and Software Engineering

EXAMINATION

Mid-year Examinations

SENG201 (C) Software Engineering I

Examination Duration: 120 minutes

Exam Conditions:

Open Book exam: Students may bring in any written or printed materials.

Any scientific/graphics/basic calculator is permitted.

Materials Permitted in the Exam Venue:

Open Book exam: Students may bring in any written or printed materials.

Materials to be Supplied to Students:

1 x Write-on question paper/answer book

Instructions to Students:

Mark or write in the spaces allocated to each answer. Do not write close to the margins, as the answer books will be scanned, and writing very close to the margin may not be picked up. If you require extra room there are space at the end of this booklet.

You should clearly indicate in the appropriate space that the answer is continued/provided elsewhere.

For Examiner Use Only

Question	Mark

Total _____

Questions Start on Page 3

1. [8 marks for whole question] Do the following Java code fragments compile? If they do not compile, explain why a fragment does not compile. If they do compile, what is their output? Note that we expect the exact output (e.g., if the output is a string then the printed string needs to be exact, including punctuation). Assume that fragments are taken from within a method of a Java class. Also, you can assume that all relevant classes are imported. (a) to (d) are not related (i.e., a variable in one question is independent from the same variable in another question).

   a. [2 marks]

```
while (true || 5/0 > 1) {
    System.out.println("I'm not sure of this one!");
      break;
}
```

   Compiles: Yes/no (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   b. [2 marks]

```
Integer try = 0;
do {
      double rand = Math.random();
if (rand * 10 < 5) {
    System.out.println("You won!");
} else {
    System.out.println("Try again.");
}
      try++;
}
while (try < 2);
```

   Compiles: Yes/no (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   c. [2 marks]

```
String s1 = "true";
String s2 = "TRUE";
Boolean s3 = s2 == s1;
System.out.println(s2 && s3);
```

   Compiles: Yes/no (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   d. [2 marks]

```
char nine = '\uffff';
char three = '\uffff';
System.out.println(nine/three);
```

   Compiles: Yes/no (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

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2. **[16 marks for whole question]** Study the following program (line numbers are provided for your convenience; they are not part of the program). You can assume that class `Flight` is implemented in file `Flight.java`

```
 1  import java.util.ArrayList;
 2  import java.util.HashSet;
 3  import java.util.List;
 4  import java.util.Set;
 5
 6  public class SomeStuff {
 7
 8      public static void main(String[] args) {
 9
10          List<Flight> listA = new ArrayList<Flight>();
11          List<Integer> listB = new ArrayList<Integer>();
12          Set<Integer> setA = new HashSet<Integer>();
13          Set<Flight> setB = new HashSet<Flight>();
14
15          listA.add(new Flight("Mexicana", "MX145", 98));
16          listA.add(new Flight("Lufthansa", "LH492", 86));
17          listA.add(new Flight("Aeroflot", "SU001", 66));
18          listA.add(new Flight("Aeromexico", "AM341", 73));
19          listA.add(new Flight("Air New Zealand", "NZ210", 98));
20
21          listB.add(5784);
22          listB.add(5487);
23          listB.add(null);
24          listB.add(7854);
25          listB.add(8457);
26          listB.add(4587);
27          listB.add(5487);
28          listB.add(7845);
29          listB.remove(1);
30          listB.remove(1);
31
32          setA.addAll(listB);
33          setA.remove(5);
34
35          setB.addAll(listA);
36          setB.add(new Flight("Jetstar", "JQ225", 60));
37
38          System.out.println("List A: " + listA);
39          System.out.println("List A size is: " + listA.size());
40          System.out.println("List B: " + listB);
41          System.out.println("Does List B contain null values? " + listB.contains(null));
42          System.out.println("Set A: " + setA);
43          System.out.println("Does Set A contain value 5487? " + setA.contains(5487));
44          System.out.println("Set B: " + setB);
45          System.out.println("Set B size is: " + setB.size());
46      }
47  }
```

a. [4 marks] State clearly for lines 11 and 12 which interface is being used and which class implements which interface in lines 11 and 12?

Your answer:

In line 11, class _____ implements interface _____.

In line 12, interface _____ is being implemented by class _____.

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b. [8 marks] Summarize the differences between the collections defined in lines 10 and 13 by indicating Yes or No to the questions provided in the table below.

Your answer:

	listA	setB
Allows positional access		
Allows duplicates		
Order of elements remains the same over time		
Allow null values		

c. [4 marks] State what the output obtained from lines 39, 41, 43 and 45 is.

Your answer:

Line 39: _____

Line 41: _____

Line 43: _____

Line 45: _____

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3. **[25 marks for whole question]** Figure 1 shows a class diagram for a "Space Explorer" game, similar to the game you developed for this year's SENG201 project. You can assume that all classes in the diagram have properties and methods necessary for the app to function but were excluded from Figure 1 for simplicity.

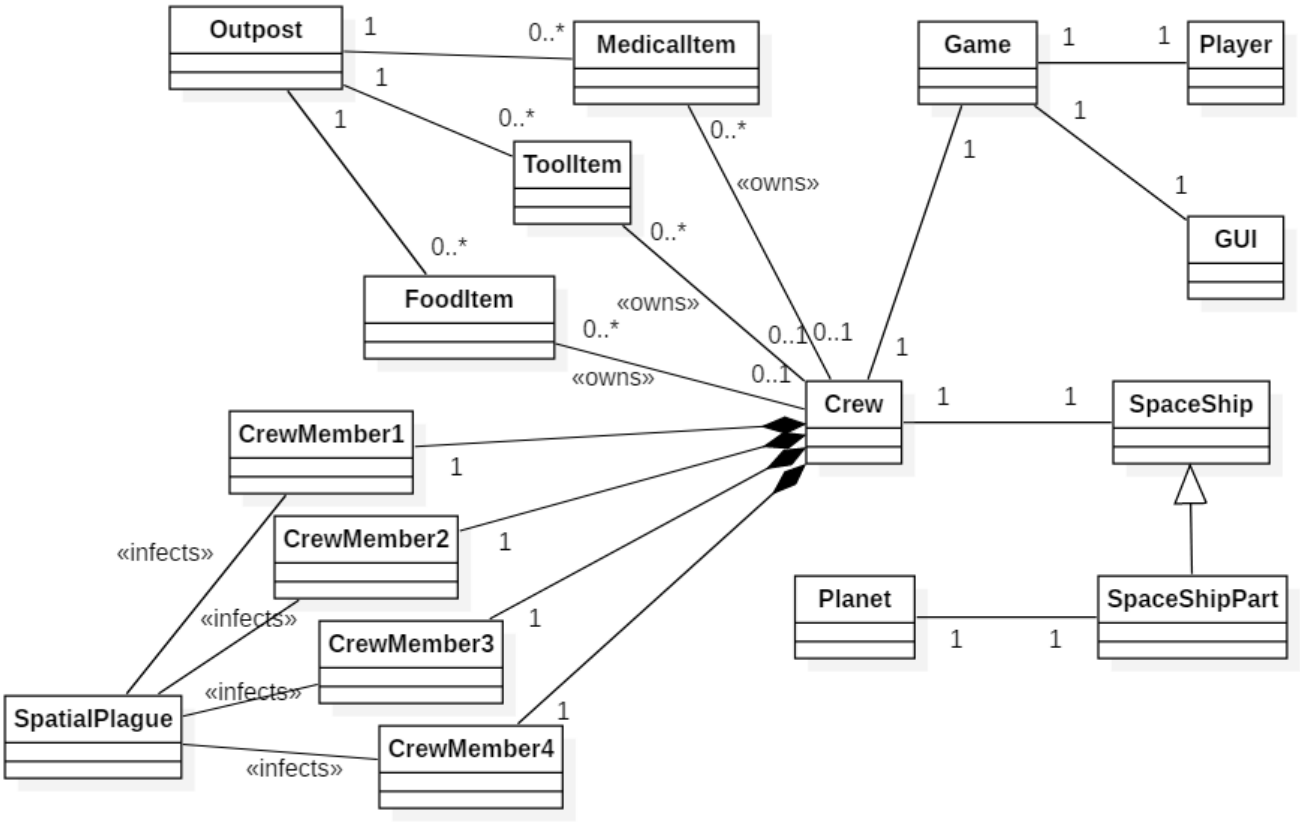


Figure 1. UML class diagram for Space Explorer game (omitting properties and methods)

Answer the following questions, and state all of the assumptions you make for your answers.

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a. [8 marks] The class design shown in Figure 1 violates several good software engineering design principles. Identify and explain two design flaws that are being violated in the class diagram and any problem(s) each flaw may cause. The design flaws must be clearly identified and explained in two or three sentences each, i.e., it is not sufficient to state "The system design is rigid". The explanation must also point out which elements of the diagram contribute to the design flaw and the problem the flaw causes.

Your answer:

b. [6 marks] Propose and explain two design patterns to correct the violations identified in question 3.a.

Your answer:

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c. [11 marks] Using the two design patterns identified in (b), redraw the redesigned UML class diagram. Your redesigned UML class diagram should include at least three different types of relationships.

Your answer:

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4. **[28 marks for whole question]** You are hired by a local software start-up to help develop a new app that is an app of apps - a super app - that acts like your own efficient travel executive assistant.
    a. [14 marks] Sketch a UML use case diagram to discuss the features of this new app with your colleagues. The app should:
        - Allow users to create their own profile; search for travel buddies with similar interests; plan their trips; post their adventures and be able to attach images, media, and text (like a mini blog); display their trips on a map; and, share them with others.
        - Eventually, allow property owners to post accommodation offers and event organizers to list their events.
        - In this way, users will have a wide variety of accommodation and leisure alternatives and they will be able to book and pay for the services (accommodation and events).

    Your use case diagram should include at least four actors, at least four use cases, and must follow UML use case diagrams notation.

Your answer:

Student ID _____
Student Number |__|__|__|__|__|__|__|__|

b. [6 marks] Draw a state transition diagram for the life cycle of a hotel reservation. The diagram should include at least three states. State all assumptions you make.

Your answer:

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Student Number |__|__|__|__|__|__|__|__|

c. [8 marks] Draft four unit test cases derived from the question 4.b diagram. Include pre and post-conditions of the test cases. Note that your test cases do not have to be fully formed JUnit tests.
Hint: An expected test case might look like this:

GIVEN: status = OFF //preconditions, the initial state
WHEN: power() //what caused the transition
THEN: status = ON //post-conditions, the expected state

Your answer:

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Student Number |__|__|__|__|__|__|__|__|

5. **[14 marks for whole question]** Figure 2 shows a sequence diagram for a travel planner app. Assume that this app follows a 3-tier architecture and all classes in the diagram have properties and methods necessary for the app to function but were excluded from Figure 2 for simplicity.

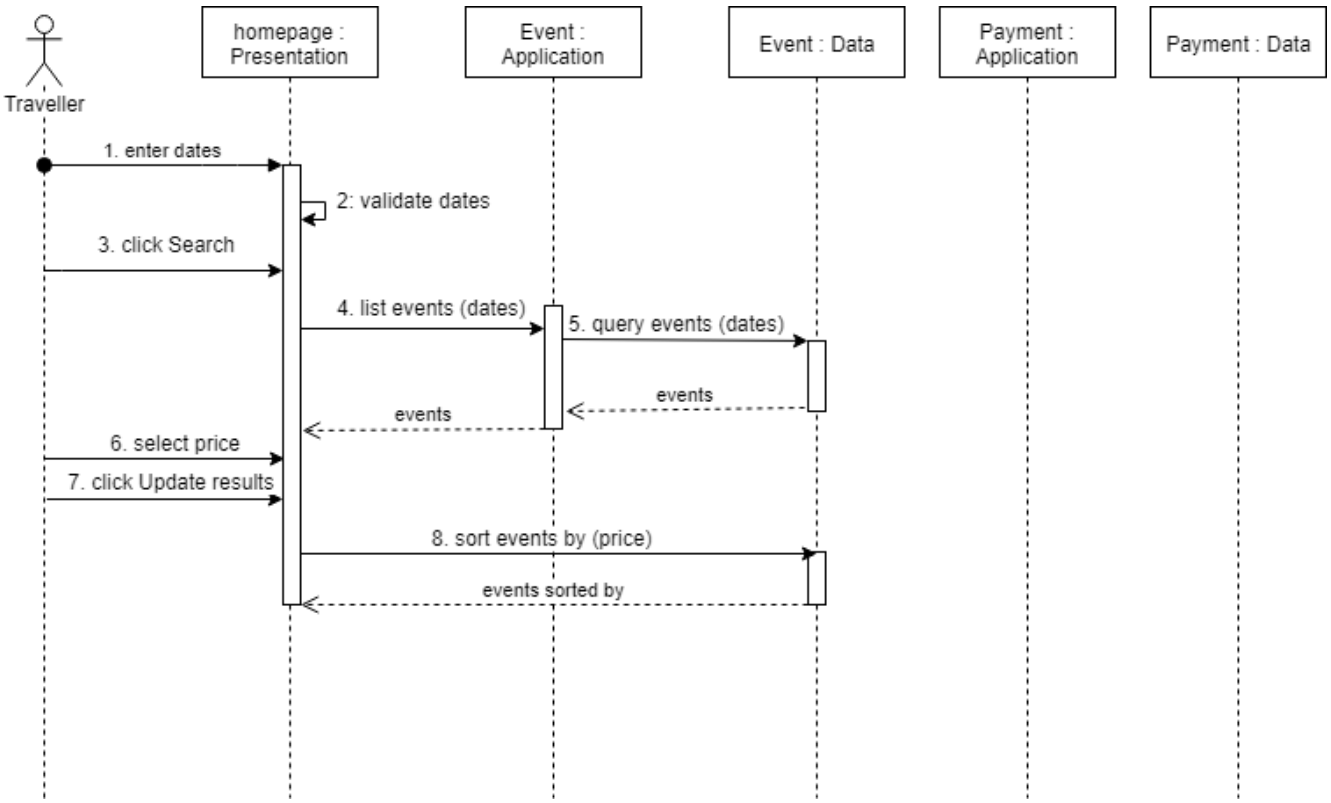


Figure 2. UML sequence diagram for a travel planner app

Answer the following questions, and state all of the assumptions you make for your answers.

a. [2 marks] Explain in your own words what the 3-tier architecture pattern is. In your answer, you should provide two different characteristics of the pattern explained as a bullet point using one or two sentences.

Your answer:

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b. [6 marks] Based on the sequence diagram in Figure 2, create a new version by:
   - Correcting errors in the diagram, if any.
   - Adding steps to it. The steps should represent an *alt* or a *loop* flow (choose only one).

   State all assumptions you make.

Your answer:

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Student Number |__|__|__|__|__|__|__|__|

c. [6 marks] Create a UML package diagram that represents the 3-tier architecture derived from the question 5.b diagram. The diagram should include the content of each package.

Your answer:

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Student Number |__|__|__|__|__|__|__|__|

6. [9 marks for whole question] List and explain in bullet points three reasons for why a risk assessment early in a software project (and throughout a project) can potentially reduce rework and increase the success rate of a software project. Include examples of risks to justify your reasons (one example per bullet).

Your answer:
Reason 1:

Reason 2:

Reason 3:

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You can use this extra page for answering any exam questions that need extra space. State clearly which question you are answering (print the number of the question).

Your answer to question ___:

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You can use this extra page for answering any exam questions that need extra space. State clearly which question you are answering (print the number of the question).

Your answer to question ___:

Student ID _____
Student Number |__|__|__|__|__|__|__|__|

End of Examination